

Surgery for lung cancer

Surgery for lung cancer removes the cancer and some lymph nodes in the chest. Operations used may include a lobectomy, pneumonectomy or wedge resection.

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Lung cancer surgery

Some people will be able to have surgery to remove the cancer. Whether surgery is possible will depend on:

- the [type of lung cancer](#)
- its [stage](#)
- your general health.

Your doctor will talk to you about whether surgery is possible in your situation. They will discuss the possible risks and benefits of surgery for you.

Lung cancer surgery is a big operation, and you need to be well enough to cope with it. Before you have surgery, your doctor will arrange tests to check how well your lungs and heart are working.

Knowing you are to have a big operation can be [worrying](#). It can help to talk about how you feel and about any concerns you have with someone from the cancer team.

A surgeon who is an expert in lung surgery will do the operation. As well as removing the cancer, the surgeon will remove an area of normal tissue around the cancer. They will also remove some of the [lymph nodes](#) in the chest.

When is surgery used for lung cancer?

This often depends on the [stage of lung cancer](#) you have. Your doctor may offer you surgery if you have a [non-small cell lung cancer \(NSCLC\)](#) stage 1 or 2.

Some people with a stage 3 non-small cell lung cancer may also be able to have surgery.

You may have other treatment before or after surgery. This may be with [chemotherapy](#), [immunotherapy](#), [targeted therapy](#) or [radiotherapy](#).

Some people may be offered a combination of chemotherapy and immunotherapy before surgery.

If the cancer is a stage 3a, you may have combined chemotherapy and radiotherapy before surgery. This is called chemoradiation. You will have the operation 3 to 5 weeks after chemoradiation, to allow you time to recover before surgery.

Surgery is rarely possible for [small cell lung cancer \(SCLC\)](#), because it has often spread outside the lung when it is diagnosed.

Booklets and resources



Booklets

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Surgery for lung cancer

Types of surgery for lung cancer

The surgery you have will depend on the size of the cancer and exactly where it is in the lungs. There are 3 main types of operation used to remove lung cancer.

Removing a lobe of the lung (lobectomy)

Your lungs are divided into lobes. The left lung has 2 lobes and the right lung has 3 lobes.

During a lobectomy, the surgeon removes one of the lobes of the lung. It is the most common operation for lung cancer. After a lobe has been removed, the remaining lobes expand to fill the space.

Image: This diagram shows the 2 lungs in the chest. Between the top part of the lungs is a tube called the windpipe. This divides into 2 smaller tubes going into each lung. Underneath the lungs is a sheet of muscle called the diaphragm. The lung on the left of the picture has a dotted area around the upper part of the lung. This shows where the upper part of the lung has been removed with an operation called a lobectomy.

If the tumour has spread to 2 lobes next to each other, the surgeon may remove 2 lobes. This is sometimes called a bilobectomy.

Removing a small part of the lung (wedge resection)

If the cancer is very small and positioned towards the edge of the lungs, the surgeon may be able to take out a smaller amount of lung tissue.

A wedge resection is where the surgeon removes the cancer and a wedge-shaped area of tissue that surrounds it.

Image: This diagram shows the 2 lungs in the chest. Between the top part of the lungs is a tube called the windpipe. This divides into 2 smaller tubes going into each lung. Underneath the lungs is a sheet of muscle called the diaphragm. The lung on the right of the picture shows a small dotted triangular area near the top of the lung. This shows where a part of the lung has been removed with an operation called a wedge resection.

A segmentectomy is where the surgeon removes a slighter larger area of the lobe. Both a wedge resection and a segmentectomy are also known as a sub-lobar resection.

A surgeon may also suggest this type of operation if there is too much damage to the lung for you to be able to have a lobectomy.

Another operation that removes part of a lung is called a sleeve resection. This type of surgery is less common. It may be done if the cancer is nearer the centre of the lungs or involves one of the bronchi, the main airways to the lungs.

Removing all the lung (pneumonectomy)

Some people may need to have a whole lung removed. This operation is called a pneumonectomy. It is less common than other types of lung surgery. It may be done when:

- the cancer is near where the airways enter the lung
- more than 1 lobe of the lung is affected.

You can still breathe normally with only 1 lung. But you might find some tasks harder to do. Your doctor will discuss the possible effects of having this type of surgery before you agree to have it.

Image: This diagram shows the 2 lungs in the chest. Between the top part of the lungs is a tube called the windpipe. This divides into 2 smaller tubes going into each lung. Underneath the lungs is a sheet of muscle called the diaphragm. The lung on the right of the picture has a dotted outline showing that a whole lung has been removed with an operation called a pneumonectomy.

Removing the lymph nodes

During surgery to remove the cancer, the surgeon also removes some of the [lymph nodes](#) close to the cancer. These are examined under a microscope to check for cancer cells. This will give your doctor more detailed information about the stage of the cancer and will help them know if you need further treatment.

How surgery for lung cancer is done

There are different types of surgery for lung cancer. Your surgeon will explain which one is most suitable for you. Before the operation you will sign a [consent form](#) agreeing to the operation.

Keyhole surgery

With keyhole surgery, the surgeon makes several small cuts instead of a single large cut. This leaves much smaller wounds, so you usually recover faster.

Sometimes the surgeon needs to change from keyhole surgery to open surgery. They will only do this if this will be better for you. Your surgeon will discuss this with you before you sign the consent form for the operation.

Video-assisted thoracoscopic surgery (VATS)



Video-assisted thoracoscopic surgery (VATS) is a type of keyhole surgery. It is done by surgeons who are specially trained in this technique.

The surgeon makes 1 or more small cuts in the skin. They put a tube with a light and video camera attached into the chest through one of the cuts. This is called a thoracoscope. The camera sends images of the inside of the chest to a computer screen.

The surgeon then passes small surgical instruments through the other cuts to remove the cancer.

After VATS you have much smaller scars than with open surgery. You may have less pain and recover faster than with open surgery. Your stay in hospital is usually shorter.

Your doctor and nurse will tell you what to expect after VATS surgery.

Robotic-assisted thoracic surgery (RATS)



Robotic-assisted thoracic surgery is a type of keyhole surgery. It is done by surgeons who are specially trained in this technique.

The surgeon makes 1 or more small cuts in the skin. They put a tube with a light and video camera attached into the chest through one of the cuts. The camera sends images of the inside of the chest to a computer screen.

Small surgical instruments are passed through the other cuts. These are attached to robotic arms. The surgeon controls the robotic arms to remove the cancer. They can move very precisely.

People often recover more quickly from this type of surgery.

Robotic-assisted thoracic surgery is not available in all hospitals. Your surgeon will tell you if it is suitable for you and where you may be able to have it done.

Open surgery

The surgeon usually makes a cut on the side of your chest. This operation is called a thoracotomy. They may move the ribs to get to the lungs. You will have a scar around the side of your chest afterwards. The scar will be 10cm to 20cm long.

Before your operation

It will help your recovery if you are as fit as possible before your operation. Once you have met the surgeon, your healthcare team may suggest things you can do to improve your general health. This is sometimes called [prehabilitation](#). It can reduce the risk of complications such as blood clots and slow-

healing wounds. It can also help you to recover faster.

Diet



Eating a healthy and varied diet can improve your well-being. You may need to gain or lose weight before your surgery. You may be given specific advice by a dietitian. We have more information on diet and cancer that you might find helpful.

Physical activity



You may be asked to do exercises to improve your heart health and make your muscles stronger. Being fitter may help reduce the risk of complications after surgery. You can improve your fitness in a few weeks by doing regular exercise, such as walking.

Stopping smoking



Try to stop smoking as soon as you can before your operation. Even stopping 2 to 3 weeks before surgery, and not smoking afterwards, reduces your risk of complications. Your wounds are likely to heal more quickly and your stay in hospital may be shorter.

It can be difficult to stop smoking, especially when you are stressed. Using the [NHS Stop Smoking Services](#) improves your chances of success. Your GP can also give you support and advice. They can provide nicotine replacement therapies on prescription.

Try to stop smoking as soon as you can before your operation. Even stopping 2 to 3 weeks before surgery, and not smoking afterwards, reduces We have more information about [stopping smoking](#).

Alcohol



If you drink alcohol it can help to cut down or stop before your treatment. Try to make sure you are staying within the [recommended guidelines](#).

Enhanced recovery programme



Most hospitals have an enhanced recovery programme for people having lung surgery. This is similar to prehabilitation. An enhanced recovery programme aims to reduce the time you spend in hospital after surgery and speed up your recovery. For example, you will be given information about diet and exercise before surgery. You may also be given nutritional supplement drinks to take before your operation.

Pre-assessment clinic



At a pre-assessment clinic, you will see a nurse who will prepare you for your operation and check you are well enough to have surgery. This usually happens 1 to 2 weeks before your operation. You may want to take someone to the appointment with you.

You usually have tests to check how well your lungs are working and your general health. These may include:

- blood and urine tests
- lung function (breathing) tests
- a chest x-ray
- a recording of your heart (ECG).

Some people have more heart tests. Your doctor will explain if you need this.

At this appointment, you can ask questions and talk about any concerns you have about the operation. The nurse may show you some simple breathing exercises to help with your recovery after surgery.

Your nurse will give you information about:

- when to come into hospital – this is usually on the day of your operation, or the day before
- how long you are likely to stay in hospital
- when you can have visitors
- what to bring to hospital with you
- when to stop eating or drinking before surgery.

The nurse may give you an antiseptic body wash to use for a few days before your operation.

They will also talk to you about [going home from hospital](#). If you live on your own or care for other people, it is important to tell your nurse. They can help you prepare for going home.

After your operation

After your operation, you will wake up in the recovery unit. From there, you usually go back to the ward. Some people may be looked after in an intensive-care or high-dependency unit for a few days.

Drips and drains

After your operation, you will probably have the following tubes in place:

A drip (infusion)



This will go into a vein in the hand or arm to give you fluids. This is removed once you are eating and drinking normally again.

A chest drain



This is a tube the surgeon puts into the chest during your surgery. This drains fluid and air into a bottle. Some people may have more than 1 chest drain. You can move around with the drain in place, carrying the bottle with you. They are usually removed after a few days.

Occasionally, people may have a small tube (catheter) put into the bladder to drain urine into a bag. This will be removed after a few days.

You may have other tubes or attachments, depending on the type of surgery you have had. These include a line going into an artery. This measures your blood pressure and oxygen levels. You may have a clip on the finger to measure your oxygen levels and monitor your pulse rate. These will be removed as soon as possible as you recover from surgery.

Getting up and moving around

The nurses and your physiotherapist will encourage you to start moving around as soon as possible. This will help you recover faster from the surgery. It will reduce the risk of complications from surgery, such as developing a [blood clot](#) or chest infection.

The nurses and your physiotherapist will help you to sit in a chair. They will encourage and help you to go for short walks as soon as you can. If you have to stay in bed, it is important to move your legs regularly. This helps your circulation and prevents blood clots.

Your breathing

A nurse may give you oxygen through a mask over your mouth and nose, or thin, soft tubes in each nostril.

A physiotherapist or nurse will show you some breathing exercises to do. Your breathing may be shallow to start with. This is normal after lung surgery. The nurse or physiotherapist will encourage you to take deep breaths. will also encourage you to cough to bring up any mucus (phlegm). This helps prevent chest infections and other possible complications.

You will have [x-rays](#) to make sure your lungs are working properly.

Pain

It is important that your [pain](#) is well-controlled to allow you to do your breathing exercises and move around. Tell your nurse or doctor if the pain is not under control. They can change or increase your painkillers, so they work better.

There are different ways to help manage pain. Your doctors may discuss with you the best way to manage pain before the operation. How your pain is managed may depend on the operation you have.

For the first few days after surgery, you may have painkillers in any of the following ways:

A nerve block



There are different types of nerve block that can be used to help with pain after lung cancer surgery. Your doctor can tell you more about nerve blocks.

Into a vein (intravenously)



This goes in your hand or arm, through a drip (infusion) or a syringe connected to a small pump. This is set to give you a continuous dose of painkillers safely. You may be able to give yourself more painkiller when you need it by pressing a button. This is called patient-controlled analgesia (PCA). Your nurses will explain this to you.

Into the space around your spinal cord (an epidural)



A surgeon puts a thin tube into your back during surgery, to deliver painkillers.

As you get better and start to eat and drink, you will start to take painkiller tablets. You can continue to take these at home. You may have some discomfort or pain in your chest for several weeks or months after surgery.

Tell your doctor or nurse about any new pain or sensations. They can check it and make sure you have the right painkillers.

Your wound

Your nurses and surgeon will check your wound regularly while you are in hospital. Tell them if:

- you have any pain or swelling
- the wound feels hot
- there is fluid leaking from the wound
- you are feeling unwell with a fever.

These are possible signs of a wound infection. Tell your nurse or doctor if you have any of these symptoms after you go home.

You usually have your stitches, clips or staples removed about 7 to 10 days after your operation. A nurse can arrange for you to have these removed when you are at home. If you have dissolvable stitches, they disappear over a few weeks.

Recovering after surgery for lung cancer

Going home

You will usually be ready to go home 3 to 7 days after your operation. You will need to arrange for someone to take you home from hospital. If you live on your own, it is a good idea to arrange to stay with someone or have someone stay with you. This could be for a few days, until you feel able to manage on your own.

Before you go home, make sure you know who to contact if you have any problems.

Recovering

It may take some weeks or months to recover, depending on the operation you have had. How long it takes to recover varies. It is faster for some people than others. Try to pace yourself and do not do too much too soon.

Your doctor and nurse will give you advice on what you can do to help your recovery. You need to avoid any heavy lifting or straining your arm on the affected side. But it is important to keep doing the exercises the physiotherapist showed you. You can slowly build up your strength and fitness with light exercise, such as short walks.

Try to eat a [healthy diet](#) as this can help your recovery. Make sure you get enough rest. If you feel tired when doing any exercise, slow down and give yourself time to recover.

You can [have sex](#) again when you feel comfortable doing so. You may need to try different positions so you are comfortable.

Contact the hospital if you have pain that is not controlled, or if you have any problems with your wound.

Driving

Your doctor will tell you when it is safe to drive after your operation. It can take about 4 to 6 weeks for you to be fit enough. You need to be able to do an emergency stop easily. At first, you may find the seatbelt presses on your wound and makes it sore. You can buy padding for seatbelts to help with this.

Some car insurance policies give specific time limits for not driving after chest surgery. Check with your insurance company.

Follow-up

You usually see your surgeon or clinical nurse specialist (CNS) a few weeks after your surgery. They will check the scar and make sure everything is healing properly. They will also give you advice about how soon you can get back to work and travel again.

They will also be able to tell you if you need to have any further treatment. This will be with an oncologist, a doctor who specialises in cancer treatments. You can also ask any questions you have.

We have more information about [what happens after surgery](#).

Getting support

Macmillan is also here to support you. If you would like to talk, you can:

- Call the Macmillan Support Line for free on [0808 808 00 00](tel:0808 808 00 00).
- [Chat to our specialists online](#)
- Visit our [lung cancer forum](#) to talk with people who have been affected by lung cancer, share your experience, and ask your questions.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our lung cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help